Research Article

Ultra-Broadband Electromagnetic Control using a Triple Circular Ring Metasurface: Surface Wave Propagation, Beam Steering, and RCS Reduction (50-100 Ghz)

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SUMMARY

This work presents a metasurface based on a triple circular ring architecture designed for multifunctional electromagnetic applications. The structure enables surface wave propagation, directional beam control, and ultra-broadband radar cross-section (RCS) reduction over the 50–100 GHz frequency range. Implemented on a cost-effective FR-4 substrate, the metasurface exhibits reflection characteristics comparable to a copper surface, with a pronounced resonance near 91 GHz. It sustains strong electric and magnetic field distributions (~1 V/m and 5×10^{-3} A/m), in contrast to conventional metallic plates. The design further demonstrates controlled radiation redirection (~67°) and stable RCS reduction (−40 to −30 dB), outperforming standard copper surfaces. These results, validated through full-wave simulations, highlight the potential of the proposed metasurface for applications in stealth technology, radar systems, and wavefront engineering.

Keywords: *Metasurface, Directional Radiation, Surface Wave Propagation, Radar Cross-Section (RCS) Reduction, Beam Steering.*

1. Unit Cell

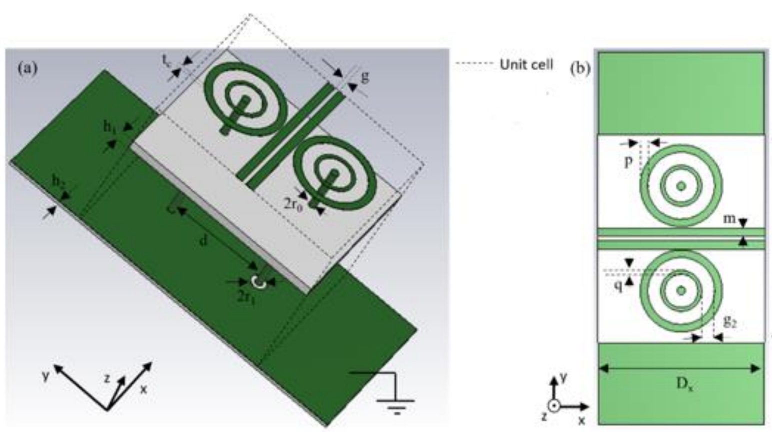


Fig.1: shows the unit cell design in different views. (a) displays a 3D view, where (b) illustrates a 2D view.

2. Results and Discussion

This paper investigates the key functionalities of the proposed metasurface, including surface wave manipulation, dynamic absorption control, directional radiation, and broadband radar cross-section (RCS) reduction. Furthermore, precise phase engineering allows effective beam steering and directional radiation. By exploiting destructive interference of reflected waves, the metasurface achieves significant monostatic RCS reduction over a wide frequency range (50–100 GHz).

a) Strong Surface Wave Propagation and Temporal Absorption

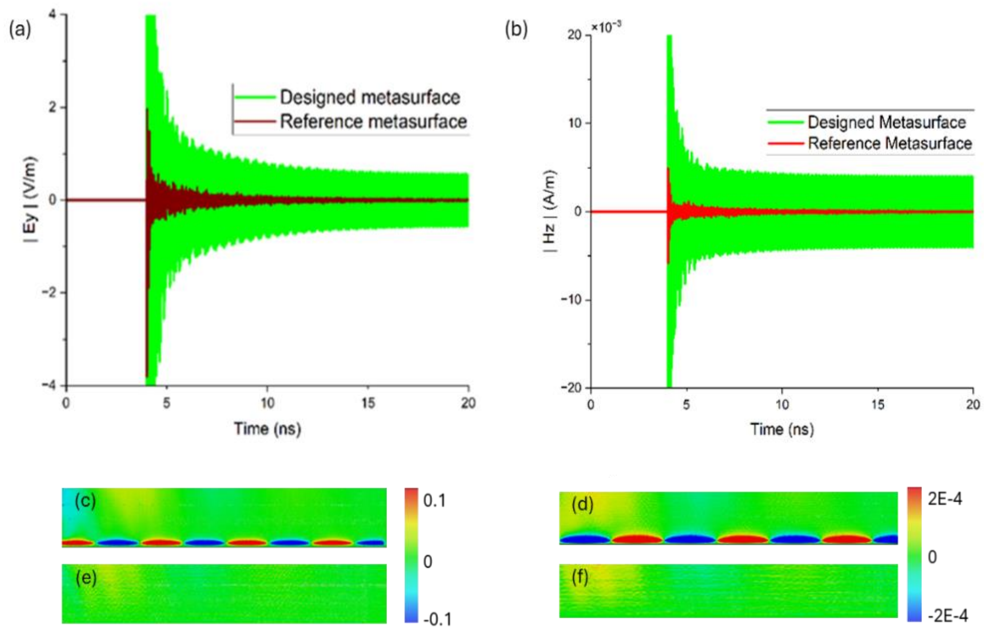


Fig. 2: Illustrates how strong the electric field (y-component) and magnetic field (z-component) are distributed compared to the reference (copper) metasurface according to the e time variation along the x-axis.

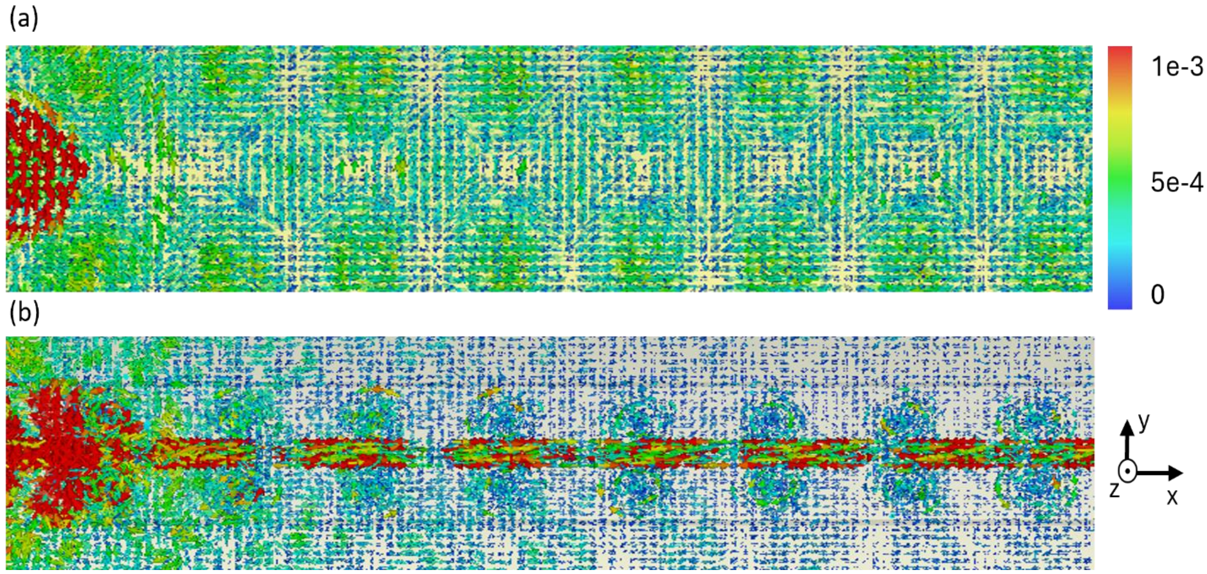


Fig.3: Illustrates the Surface current distribution (A/m) of both (a) reference (copper) metasurface and (b) our designed metasurface at time t = 5 ns.

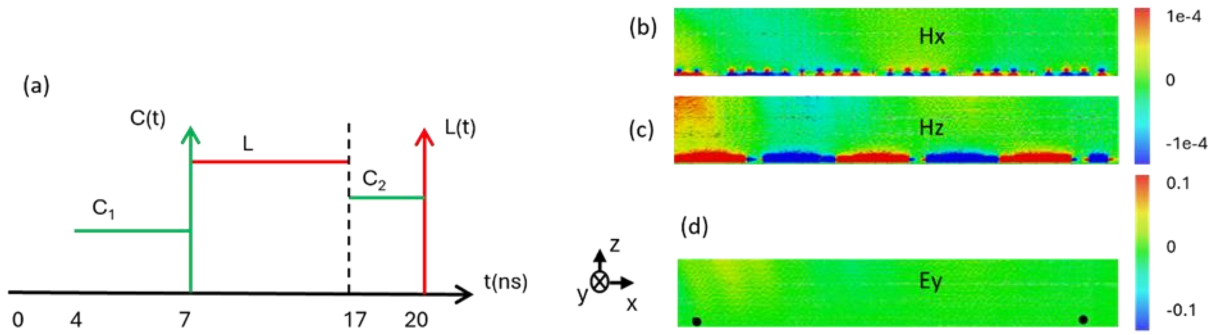
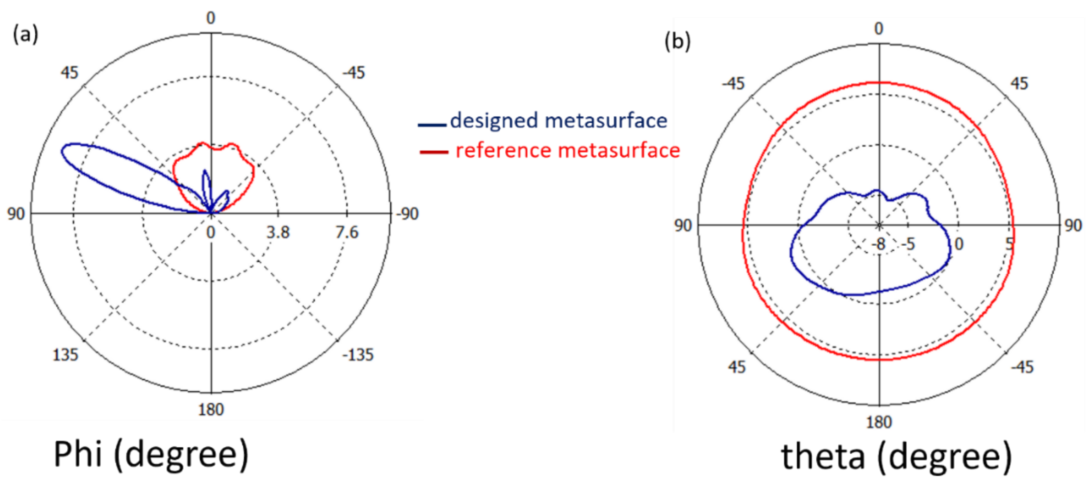


Fig. 4:(a) illustrates how the boundary conditions evolve over the entire simulation timeframe. Figures (b)–(d) display the two-dimensional distributions of the electric and magnetic fields on the designed metasurface during the field-freezing interval (7 ns < t < 17 ns). In Figures (b) and (c), the magnetic field components Hx and Hz exactly align with theoretical expectations. Furthermore, Figure (d) verifies the complete suppression of the electric field during the frozen state.

b) **Enhanced Directional Radiation:**



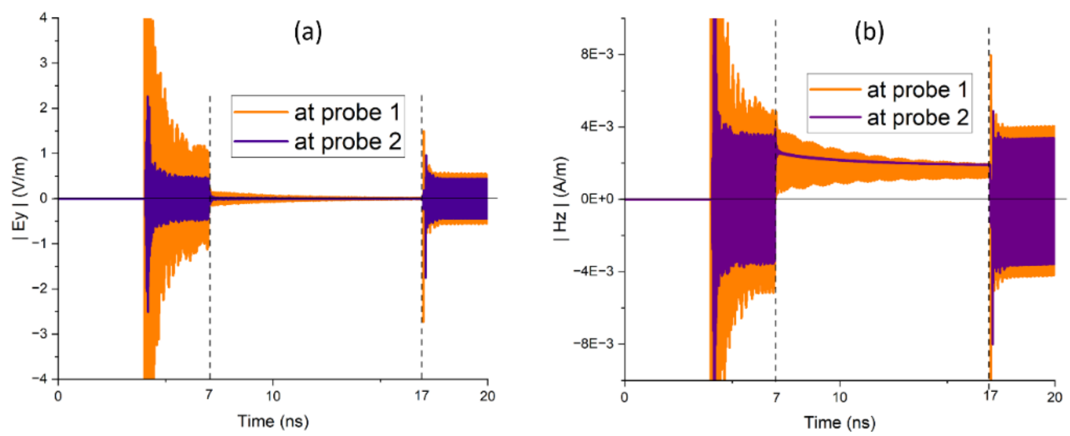


Fig. 5: illustrates the far-field directivity results under normal plane wave incidence on the metasurfaces.
(a) Shows a polar plot of the phi angle at 3.842 GHz. (b) Presents a polar plot of the theta angle at the same frequency.

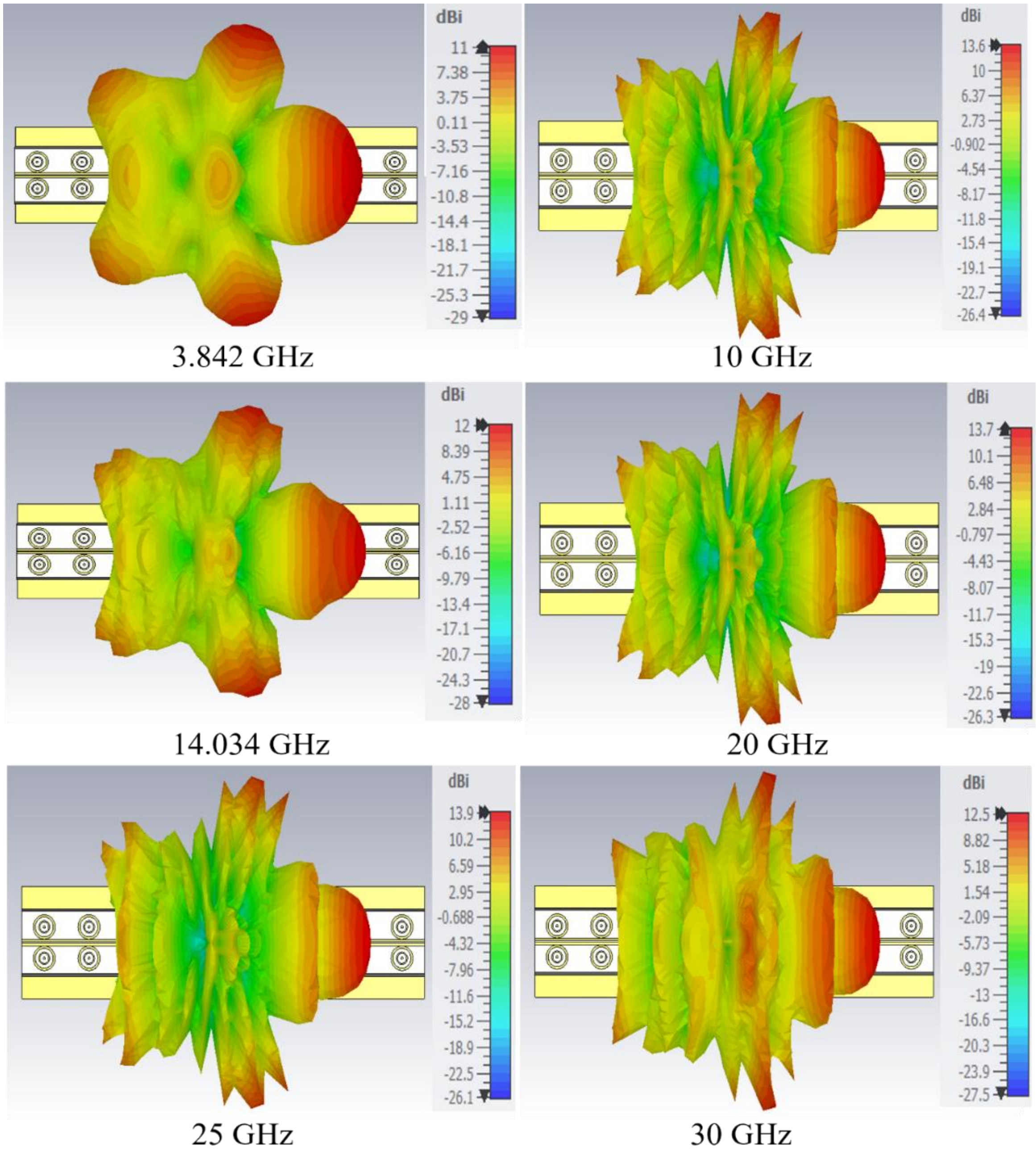


Fig. 6: presents the 3D directivity pattern of the designed metasurface at different GHz.

c) Ultra-broadband RCS Reduction for high frequencies:

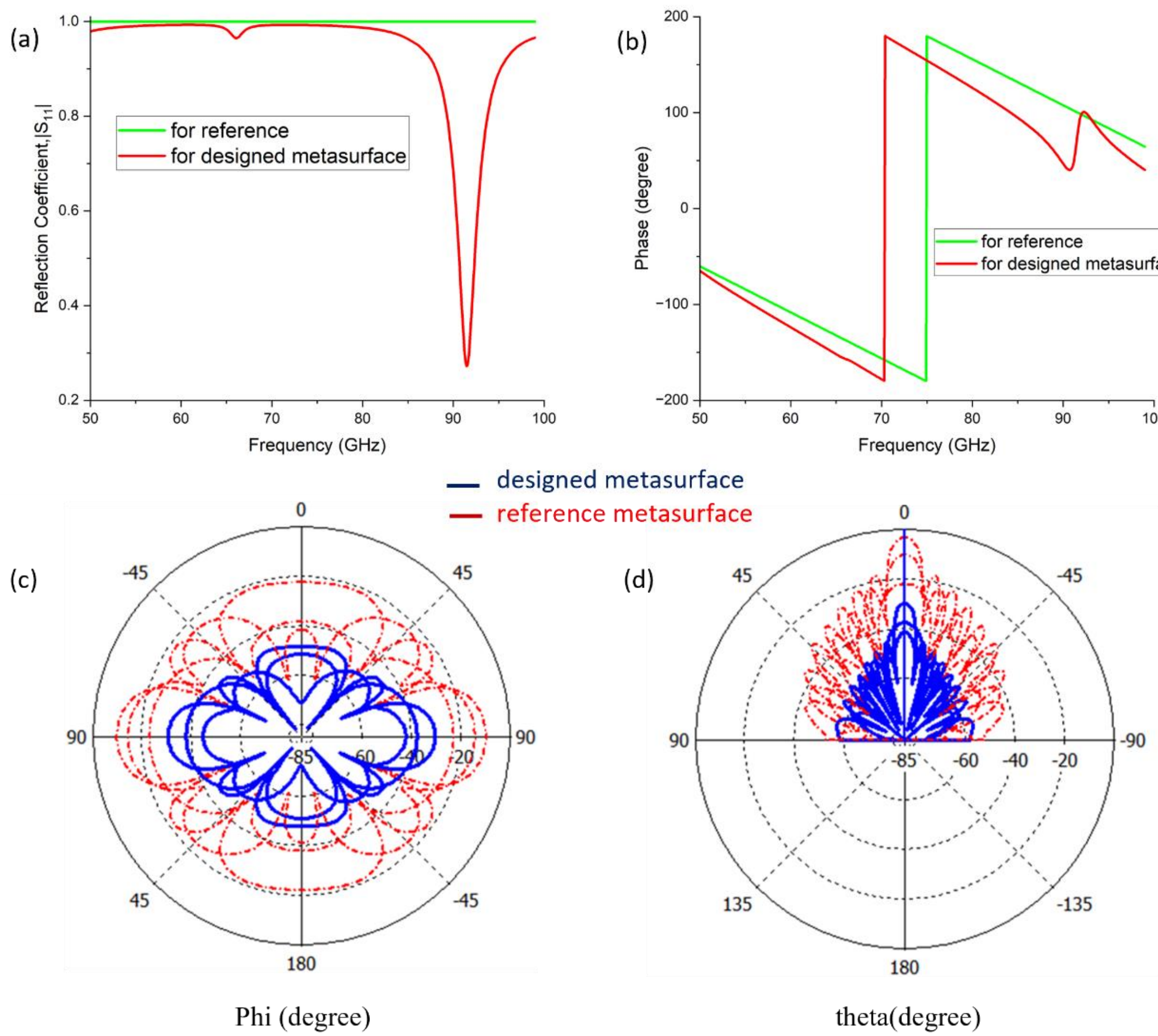


Fig.7: Farfield results. (a) and (b) show the reflection amplitude and phase of the unit cell, respectively. (c) and (d) display polar graphs of bistatic scattering in the range of -85 to 0 dB at frequencies like 50, 78, and 99 GHz.

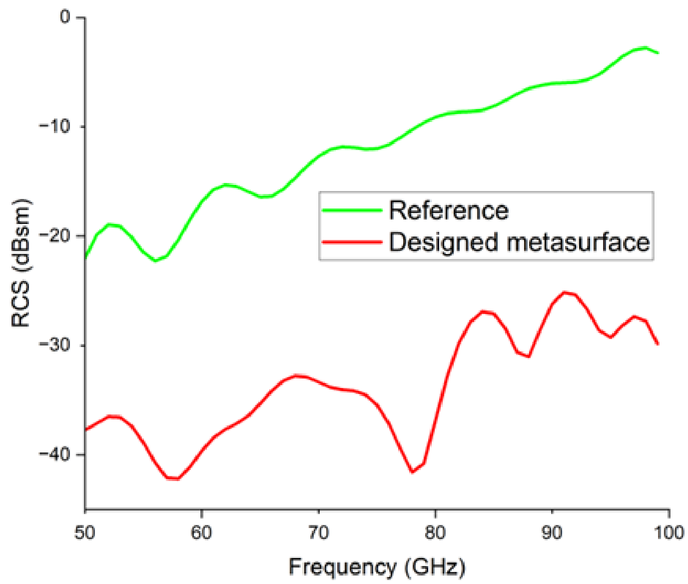


Fig. 8: shows a monostatic RCS graph of the designed metasurface, which is significantly lower compared to the reference (copper) surface with such a wide bandwidth of 50 GHz.

3. Conclusion

This work proposes a novel metasurface design based on a triple circular ring architecture for advanced electromagnetic wave manipulation. The structure supports multiple functionalities, including efficient surface wave transmission, time-dependent absorption, directional beam control, and ultra-broadband radar cross-section (RCS) reduction. Implemented on an FR-4 substrate, the metasurface demonstrates enhanced electric and magnetic field confinement, enabling improved energy localization. A key feature is its dynamic surface reactance behavior, allowing adaptive electromagnetic response within a controlled framework. The design also enables effective redirection of scattered fields, resulting in consistently low RCS over a wide frequency range. These characteristics make the proposed metasurface highly suitable for applications in electromagnetic stealth, radar systems, and wavefront engineering.

References

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